

Matthew Berger

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Senior Staff Rust Software Engineer working with Rust since 2016, with 5 years in production robotics. Founding software engineer at Hyphen, where in the first 22 months I built the food assembly robotics platform from first commit to a full controls suite, underpinning \$35M+ in combined investment from Chipotle and Cava. The platform runs on a real-time async pub/sub message broker with sub-millisecond latency and zero message loss across 4+ years of fleet operation. Open source: 280k+ crates.io downloads (freecs, enum2 family) and the Nightshade game engine (live demo at matthewberger.dev/nightshade).

Work Experience

Hyphen (Culinary Robotics)

Remote

Founding Software Engineer (Senior → Staff → Senior Staff)

July 2021 – May 2026

- Architected and built the food assembly robotics controls software, with 1,000+ pull requests merged across 4 years, from getting the first motor moving to orchestrating cloud services
- Diagnosed scaling limitations in the legacy PLC system, led the technical pivot to embedded Rust, and pioneered RP2040 firmware with Embassy-rs, creating the first Rust TMC5160 motor and AS5048A encoder drivers proving Rust could meet real-time control loop requirements
- Designed and built the async pub/sub message broker over TCP with cross-host bridging, length-prefixed postcard framing, and compile-time topic contracts via procedural macros, coordinating every distributed process across the industrial PC and RP2040 boards at sub-millisecond local latency and zero message loss across 4+ years of continuous fleet operation
- Implemented a `no_std` embedded broker client on Embassy with bounded static channels, integrating RP2040 firmware into the same pub/sub topology as the industrial PC
- Authored the controls suite as a multi-process system of state-machine-based servers corresponding to the robot equipment, communicating via typed command/event contracts over the broker, with coordinator servers orchestrating cross-subsystem behavior
- Built a custom Yocto Linux distribution and AWS Greengrass-managed OTA pipeline delivering firmware and software updates to the production makeline fleet
- Shipped Hyphen Studio, a Dioxus desktop application enabling operators to configure makeline layouts and parameters directly
- Rewrote the TypeScript/GraphQL tablet portal in pure Rust with Leptos, connecting over WebSockets to a forwarding server that bridges broker topics out to web clients, enabling direct import of the same Rust contract types used across the controls suite
- Built Hyphen Explorer (Bevy + egui) for real-time IPC visualization, debugging, and system control, achieving 100% adoption across all Hyphen engineers
- Mentored 6 engineers new to Rust into productive contributors within 2 months each, enabling parallel horizontal platform expansion over the following year

Sierra Nevada Corporation

Englewood, CO

Software Engineer III

May 2020 – July 2021

- Developed C++/Rust aerospace imaging system capable of collecting and orthorectifying 5GB/sec of pixel data during a flight
- Built asynchronous Rust simulator for unavailable flight hardware, saving 3 months on project timeline

Scientific Games

Software Engineer

Reno, NV

July 2019 – May 2020

- Contributed to the GDK, Scientific Games' Unity-based engine powering casino game titles across their portfolio
- Migrated multiple shipped casino game titles from Unity's legacy GUI text system to TextMeshPro, modernizing typography rendering across the GDK portfolio

Hamilton Company

Software Engineer (Intern through Full-Time)

Reno, NV

October 2014 – July 2019

- Built safety-critical software in a cross-disciplinary environment that calibrates and operates liquid-handling medical robots
- Reduced calibration development from 2 months to 2 weeks by consolidating multiple applications into a single GUI and reusable plugin framework
- Developed the calibration application that a major customer, Illumina (\$2M in machine purchases), used to drive Hamilton's robot firmware
- Built software adapters bridging SiLA-compliant and non-SiLA lab instruments, expanding Hamilton's interoperability across third-party device manufacturers and driving instrument sales

Open Source

- **Nightshade**: data-oriented Rust game engine (200k+ LOC) featuring a custom render graph with sub-graph composition and resource aliasing for transient memory reuse, built-in editor, rapier3d physics, kira audio, gltf asset loading, scene graph, object picking, and Steam integration. Runs natively on DX12/Metal/Vulkan and in-browser via WebGPU (live demo: matthewberger.dev/nightshade)
- **freecs** (48k+ downloads on crates.io): archetype-based ECS in Rust with zero unsafe code, Rayon-parallel system execution, sparse-set tags, deferred command buffers, and watermark change detection. The entire ECS is generated at compile time via a declarative macro. Design documented in a 3-part series at matthewberger.dev/articles
- **enum2 macro family** (enum2str, enum2contract, enum2egui, enum2pos, enum2repr): derive macros for enum-driven design patterns with 230k+ total downloads on crates.io
- **bamboo-ssg**: fast Rust static site generator with markdown + Tera templates, syntect-highlighted code blocks, live-reload dev server with incremental builds, RSS/Atom/sitemap, and responsive images. Powers matthewberger.dev
- **cameras**: cross-platform Rust library for real-time video frame pipelines across Linux, macOS, and Windows, with data-oriented design (explicit format negotiation, push-based frame delivery, zero trait objects). Ships with Dioxus and egui integrations.
- **scoop-nerd-fonts**: maintainer of the canonical scoop bucket providing one-command font installation on Windows for Nerd Fonts and other open fonts (440+ stars)

Writing

- **Build your own ECS** (3-part series at matthewberger.dev/articles): builds the freecs kernel from scratch, covering archetype storage, structural change and queries, change detection, events, tags, and command buffers

Skills

Languages: Rust, C#, C++, TypeScript, Python

Embedded & Real-time: RP2040, Embassy, `no_std`, real-time, low-latency systems, async Rust (Tokio), procedural macros, Yocto Linux

Distributed Systems: message brokers, HTTP/WebSocket, postcard, tracing/observability

Cross-Platform: Linux, macOS, Windows, Android, WebAssembly, Steam Deck, OpenXR/VR

UI/Graphics: Leptos, Dioxus, egui, Bevy, wgpu, Vulkan, DX12, Metal, OpenGL, PBR rendering

Cloud/Infra: AWS, Pulumi, Greengrass IoT, Docker, GitHub Actions

Education

University of Nevada, Reno

2017

B.S. Computer Science & Engineering, Minor in Mathematics